日本語論文英訳

[指示]
・意味が伝わる最小限の英文で構成してください。
・抽象的な表現をなくし、具体的な表現に直してから英訳してください。
・専門分野に沿った動詞の使い方をしてください。

1.まえがき

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Breast cancer is the leading cancer in Japanese women, affecting over 90,000 annually. Breast-conserving surgery (BCS) is common. For BCS, precise resection is a balance: more tissue ensures cancer removal, less preserves breast shape. Pre-operative MRI is crucial for this [1]. Conventional MRI for BCS is prone; surgery is supine. Surgeons must account for tumor displacement between positions, relying on experience. Amano et al.'s tumor projection mapping technology [2] could improve resection accuracy. This requires prone and supine contrast-enhanced MRI (CE-MRI). However, few patients opt for two CE-MRI scans due to financial and physical burdens (e.g., allergy risk from two contrast administrations).
Previous research [3][4] used CycleGAN for interconversion between CE-MRI and non-CE MRI. While successful for prone CE-MRI and prone non-CE MRI, conversion between supine non-CE MRI and supine CE-MRI failed. GANs have unstable training, creating a need for more stable, recent generative models.
This research, therefore, employs Diffusion Models [5], known for training stability, to convert between prone CE-MRI and non-CE MRI, and evaluates the converted image quality.

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Breast cancer is the most common cancer among Japanese women, with over 90,000 new cases annually. Breast-conserving surgery, which removes only part of the tissue, is widely performed. A balance is needed between removing enough tissue to prevent recurrence and preserving the breast shape. Preoperative MRI is essential for planning.
Conventional MRIs are taken in the prone position, while surgery is performed supine, creating a positional mismatch that relies on the surgeon's experience.
Amano et al. proposed a projection mapping method to accurately align tumor positions, requiring both prone and supine contrast-enhanced MRIs. However, administering contrast agents twice increases cost and health risks, reducing patient uptake.
Previous studies used CycleGAN to convert between contrast and non-contrast MRIs with a single contrast dose. While conversion was successful for prone images, it failed for supine

ones. Additionally, GANs are unstable during training.
This study explores the use of Diffusion Models, known for stable learning, to convert between prone contrast and non-contrast MRIs and evaluates the quality of the generated images.

「まえがき」について、chatGPT4o を採用
→抽象的表現がなく、「prone」「supine」など専門分野の用語が使用されているため。

2.方法

2.1 腹臥位乳房 MRI の変換に用いたモデル

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We used a Conditional Diffusion Model based on the Med-DDPM framework. Diffusion Models consist of two stages:

1. Forward (Diffusion) Process: Gradually adds Gaussian noise to an input image over a series of timesteps until it becomes pure noise.
2. Reverse (Denoising) Process: Learns to reverse the noising process, starting from pure noise and generating a clean image.

Conditional Diffusion Models guide this generation process based on a given condition. Med-DDPM was originally proposed for brain MRI synthesis, using a brain or tumor mask as the condition.
In this study, we adapted Med-DDPM for breast MRI. As a preliminary step toward our ultimate goal of transforming supine NCE-MRI to supine CE-MRI, we focused on the prone-to-prone transformation. We used the prone NCE-MRI as the conditional input to generate the corresponding prone CE-MRI.

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Diffusion Models consist of two steps: forward diffusion and reverse denoising. Gaussian noise is added to the input image over time to create a noise-only image. Then, a neural network denoises it step-by-step to recover the original.
We use Med-DDPM, a conditional diffusion model that takes an input (here, non-contrast prone MRI) and generates a corresponding contrast-enhanced version. Unlike its original version for brain MRI, we omit mask images and input the non-contrast image directly.

「腹臥位乳房 MRI の変換に用いたモデル」について、GoogleAIstudio を採用
→
・「forward diffusion」「reverse denoising」について順序立てて説明しているため。
・Gaussian noise を段階的に加えるプロセスが内容の沿っているため。
・「conditional input」と専門的用語を用いているため。

2.2 データセット

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We used a dataset of paired prone NCE-MRI and CE-MRI scans from Nihon University Hospital. The dataset was divided as shown in Table 1.As a preprocessing step, we normalized pixel values to the range [-1, 1] to reduce noise in the generated images.

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We used prone MRI pairs (contrast and non-contrast) from Nihon University Hospital:
Training: 340 pairs
Test: 10 pairs
(All images normalized to [-1, 1] to suppress noise.)

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・訓練データ 340 ペア、テストデータ 10 ペアといった数値情報が提示されているため。
・正規化の目的が簡潔に述べられているため。

2.3 評価方法

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We performed a qualitative evaluation by visually comparing the generated prone CE-MRI with the ground-truth scans.

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We qualitatively compared model outputs with true contrast MRI.

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→よりシンプルで直接的であり、簡潔にまとめられていると感じたため。

3.結果

3.1 Med-DDPM の学習方法

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We trained the Med-DDPM using three different sets of hyperparameters, as shown in Table 2. We moved from Setting 1 to Setting 2 to reduce image noise. This involved changing the loss function from L1 to Log-Cosh, which better handles small errors, and increasing the learning rate and batch size. To further improve image quality, we increased the number of training epochs from 5000 to 10000 in Setting 3.

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We modified the baseline to reduce noise:

Setting	Before	After	Purpose
Loss Function	L1	Log-Cosh	Handle small errors better
Learning Rate	1e-5	1e-4	Faster convergence
Batch Size	1	2	Smoother gradients
Epochs	5000	10000	Sharper tumor borders

Log-Cosh loss combines the advantages of L1 and L2, reducing residual noise from unstable samples.

「Md-DDPM の学習方法」について、chatGPT4o を採用

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・表形式で端的に設定変更を整理しているため。

・Log-Cosh 損失関数の利点について簡潔に述べていると感じたため。

3.2 推論結果

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Figure 1 and Figure 2 show the ground-truth NCE-MRI and CE-MRI test images, respectively. Figures 3-5 show the images generated by the models trained with Settings 1-3.

・Setting 1 (Fig. 3): Compared to the ground truth (Fig. 2), the generated image roughly captures the outlines of the breast and torso, but it is significantly noisy. This is likely because the L1 loss function struggles to reconstruct fine details.

・Setting 2 (Fig. 4): This image shows less noise than the result from Setting 1. The tumor in the left breast is enhanced, demonstrating an improvement in reproducing detailed features.

However, the overall contrast is low, and the tumor enhancement is weaker than in the ground truth.

・Setting 3 (Fig. 5): Increasing the epochs further reduced noise and produced a sharper tumor outline compared to Setting 2. This confirms the benefit of longer training. However, the tumor enhancement is still weaker than in the ground-truth image (Fig. 2).

The remaining image artifacts suggest that the hyperparameters were not fully optimized.

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・Baseline (L1): Preserved basic structure but high noise and weak contrast.

・After tuning (Log-Cosh): Reduced noise, improved tumor edge visibility.

・After 10k epochs: Sharper tumor outlines, further noise reduction.

However, the brightness of cancer regions was still weaker than the real contrast MRI. This suggests underfitting or insufficient contrast replication.

「推論結果」について、GoogleAIstudio を採用

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・各設定の結果（Setting1〜3）に対して、図を参照しているため。

・画像の輪郭やノイズについて内容を漏らさず言及しているため。

4.むすび

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In this study, we applied a Conditional Diffusion Model (Med-DDPM) to generate prone CE-MRI from prone NCE-MRI. By adjusting hyperparameters—specifically, changing the loss function to Log-Cosh and increasing the number of epochs—we successfully reduced noise and improved the reproduction of tumor outlines in the generated images.

Despite these improvements, the generated CE-MRI exhibited weaker tumor enhancement compared to the ground-truth images. This indicates a need for further refinement.

Future work will involve a systematic optimization of hyperparameters and exploring additional conditioning data to improve the quality of the generated images. We also plan to incorporate quantitative metrics, such as Root Mean Square Error (RMSE) and Structural Similarity Index (SSIM), for a more rigorous evaluation.

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We generated contrast-enhanced prone MRI from non-contrast input using a modified Med-DDPM. Adjusting loss, learning rate, batch size, and epoch count improved output quality.

Log-Cosh loss enhanced contrast structure retention. More epochs improved edge sharpness. However, tumor brightness remains insufficient. Future work includes:

・Better hyperparameter tuning

・Adding input conditions

・Evaluating with metrics such as RMSE and SSIM

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→「tumor enhancement」「ground-truth images」と専門的かつ具体的な表現を使っているため。